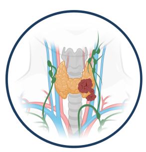
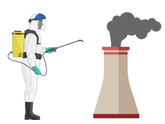
Environmental Chemical Burden in Differentiated Thyroid Cancer

Human Thyroid Tumor Exposomics

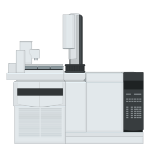
• Recent increases in differentiated thyroid cancer (DTC) incidence.



• Potential contribution of environmental chemicals to epidemiologic trends.

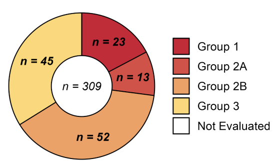


• GC–MS environmental surveillance study of 709 xenobiotic compounds in n = 60 DTC samples and n = 8 cadaver thyroids.

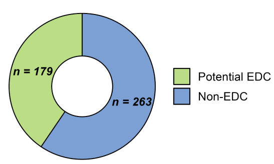


Environmental Chemical Surveillance

IARC Carcinogen Class

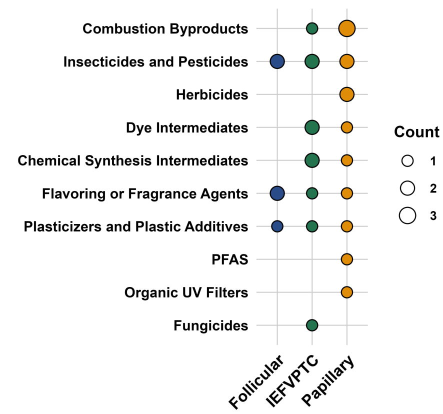


Endocrine-Disrupting Chemicals



• 442 chemicals annotated in tumors, spanning insecticides, herbicides, industrial chemicals, plasticizers, flame retardants, and flavoring or fragrance agents.
• 40% were potential EDCs and 23 were IARC Group 1 carcinogens.

Type-Specific Chemical Differences



• 29 identified chemicals differed between types; papillary thyroid carcinoma samples had higher levels of multiple combustion byproducts, insecticides and pesticides, and herbicides.

Conclusion: *This study provides evidence for accumulation of environmental chemicals in DTC; specific organic pollutants may influence type-specific carcinogenesis.*

Reference: Preston JD, Liang Y, Szabo Yamashita T, et al. Environmental Chemical Burden in Differentiated Thyroid Cancer. *Environment International*. 2026.

Figure 1

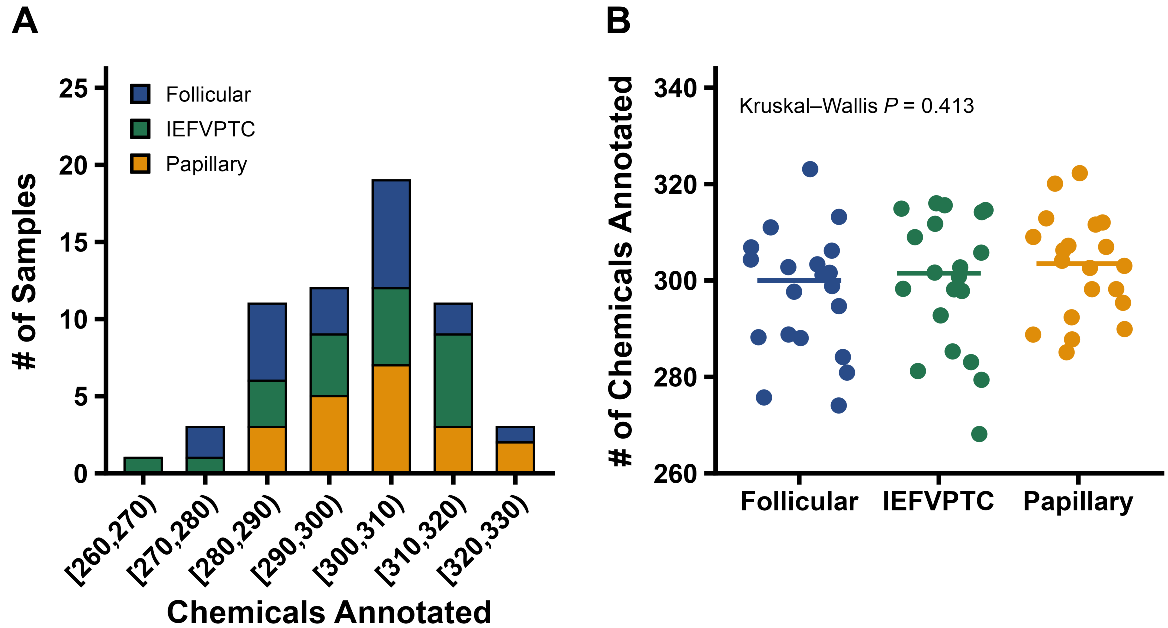
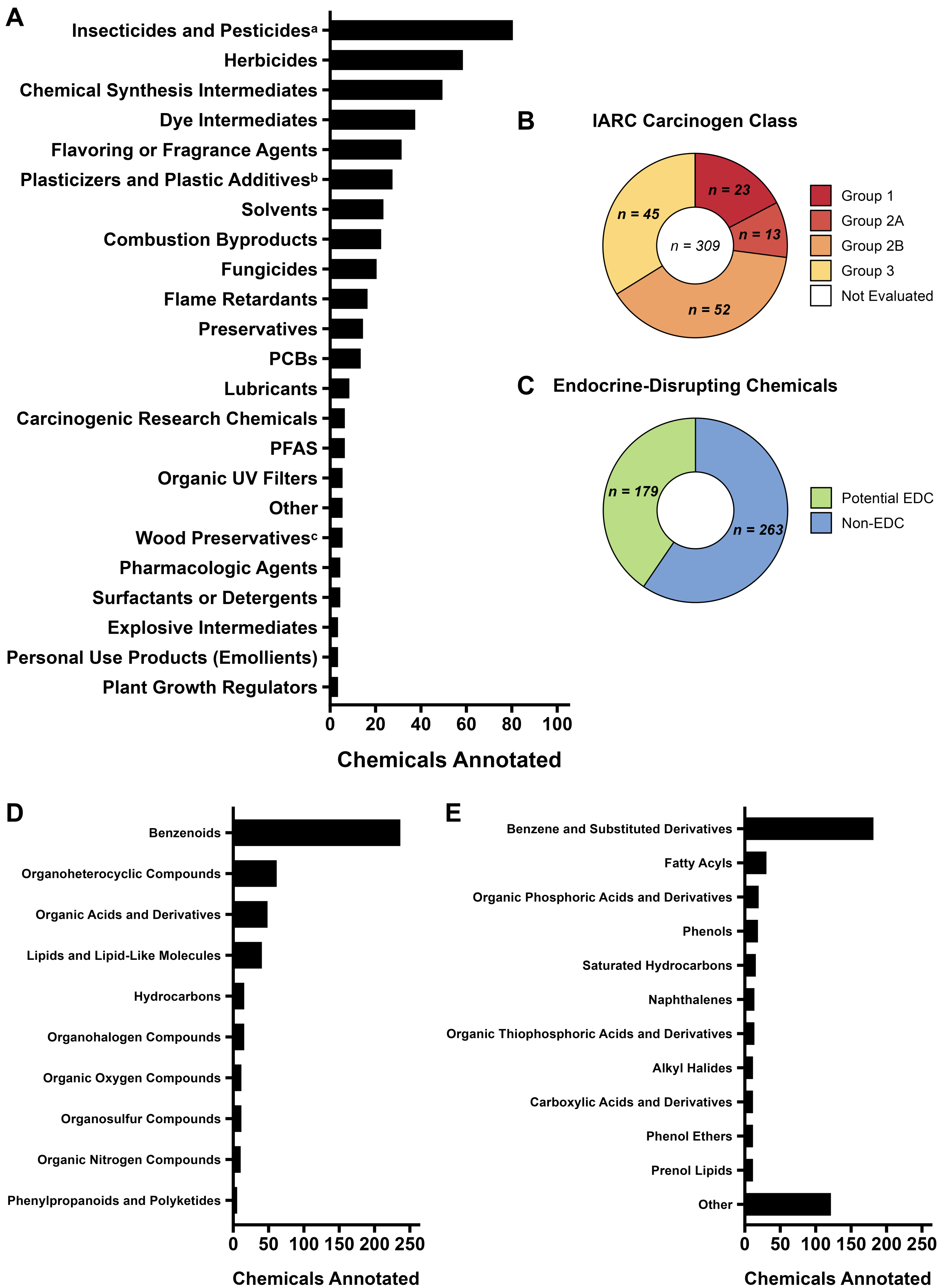
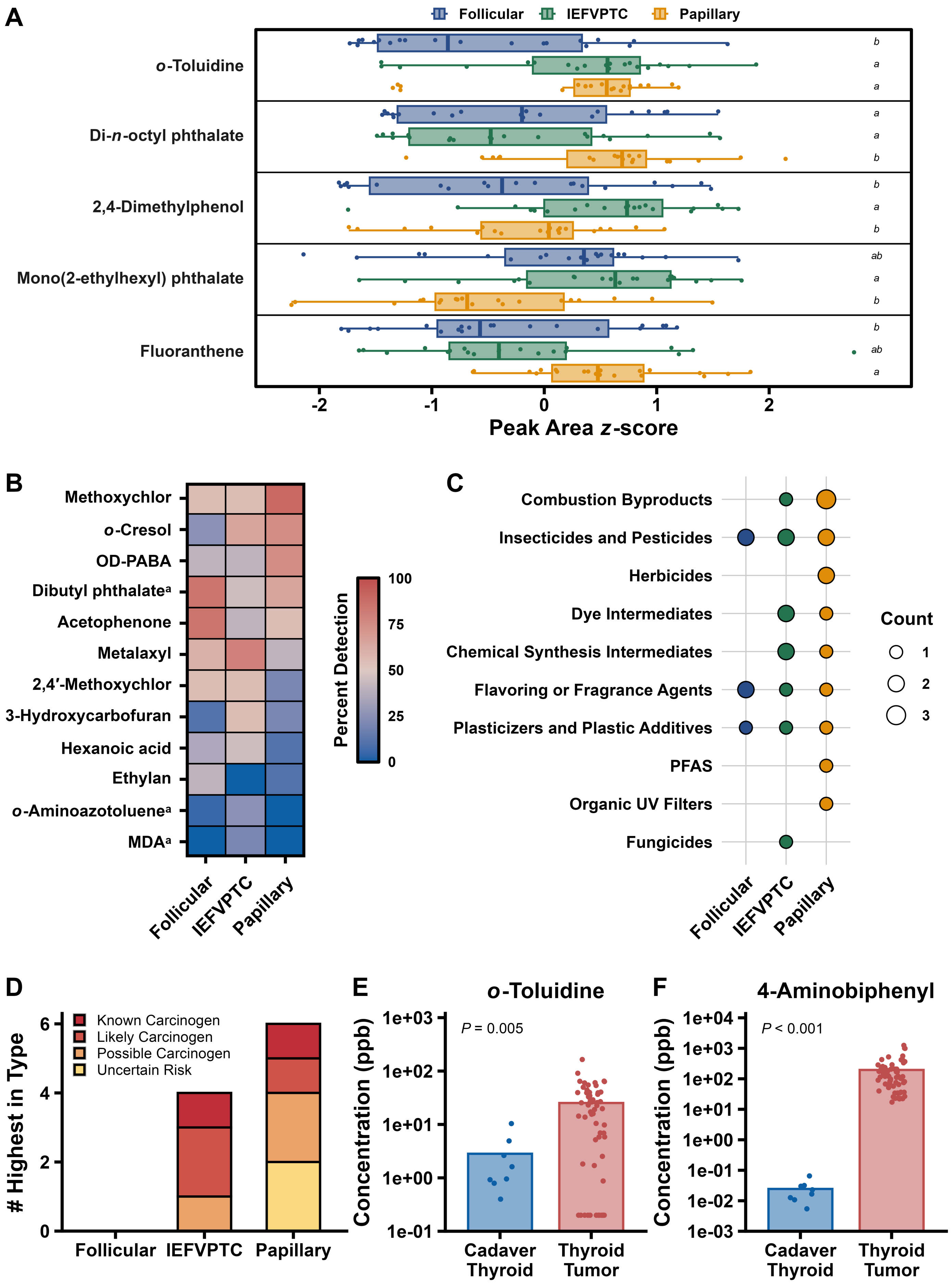

A
B
25
20
15
10
5
0
# of Samples
Follicular
IEFVPTC
Papillary
[260,270)
[270,280)
[280,290)
[290,300)
[300,310)
[310,320)
[320,330)
Chemicals Annotated
340
320
300
280
260
# of Chemicals Annotated
Kruskal–Wallis P = 0.413
Follicular
IEFVPTC
Papillary

Figure 2



A
Insecticides and Pesticides[a]
Herbicides
Chemical Synthesis Intermediates
Dye Intermediates
Flavoring or Fragrance Agents
Plasticizers and Plastic Additives[b]
Solvents
Combustion Byproducts
Fungicides
Flame Retardants
Preservatives
PCBs
Lubricants
Carcinogenic Research Chemicals
PFAS
Organic UV Filters
Other
Wood Preservatives[c]
Pharmacologic Agents
Surfactants or Detergents
Explosive Intermediates
Personal Use Products (Emollients)
Plant Growth Regulators
0 20 40 60 80 100
Chemicals Annotated
B
IARC Carcinogen Class
n = 23
n = 13
n = 52
n = 45
n = 309
Group 1
Group 2A
Group 2B
Group 3
Not Evaluated
C
Endocrine-Disrupting Chemicals
n = 179
n = 263
Potential EDC
Non-EDC
D
Benzenoids
Organoheterocyclic Compounds
Organic Acids and Derivatives
Lipids and Lipid-Like Molecules
Hydrocarbons
Organohalogen Compounds
Organic Oxygen Compounds
Organosulfur Compounds
Organic Nitrogen Compounds
Phenylpropanoids and Polyketides
0 50 100 150 200 250
Chemicals Annotated
E
Benzene and Substituted Derivatives
Fatty Acyls
Organic Phosphoric Acids and Derivatives
Phenols
Saturated Hydrocarbons
Naphthalenes
Organic Thiophosphoric Acids and Derivatives
Alkyl Halides
Carboxylic Acids and Derivatives
Phenol Ethers
Prenol Lipids
Other
0 50 100 150 200 250
Chemicals Annotated

Figure 3



A
Follicular
IEFVPTC
Papillary
o-Toluidine
Di-n-octyl phthalate
2,4-Dimethylphenol
Mono(2-ethylhexyl) phthalate
Fluoranthene
Peak Area z-score
B
Methoxychlor
o-Cresol
OD-PABA
Dibutyl phthalate[a]
Acetophenone
Metalaxyl
2,4′-Methoxychlor
3-Hydroxycarbofuran
Hexanoic acid
Ethylan
o-Aminoazotoluene[a]
MDA[a]
Percent Detection
C
Combustion Byproducts
Insecticides and Pesticides
Herbicides
Dye Intermediates
Chemical Synthesis Intermediates
Flavoring or Fragrance Agents
Plasticizers and Plastic Additives
PFAS
Organic UV Filters
Fungicides
Count
D
Known Carcinogen
Likely Carcinogen
Possible Carcinogen
Uncertain Risk
Highest in Type
E
o-Toluidine
P = 0.005
Concentration (ppb)
Cadaver Thyroid
Thyroid Tumor
F
4-Aminobiphenyl
P < 0.001